Algebra Review: Unit 4 — Graphing Inequalities on the Coordinate Plane

Station 1 of 4: Testing Solutions of Linear Inequalities

Name: _____ Date: _____

Substitute the ordered pair into the inequality to decide whether it is a solution. Answer Yes or No.

1. Is $(1, 3)$ a solution of $y > 2x - 1$?

2. Is $(3, 2)$ a solution of $y \leq -x + 4$?

3. Is $(4, 5)$ a solution of $y \geq \frac{1}{2}x + 3$?

4. Is $(3, 2)$ a solution of $2x + 3y < 12$?

5. Is $(7, 3)$ a solution of $x - y \geq 4$?

6. Is $(2, -3)$ a solution of $y < -3$?

7. Is $(-1, -5)$ a solution of $3x - 2y > 5$?

8. Which of the points $(0, 0)$, $(3, -1)$, and $(-2, 2)$ are solutions of $y > -x$?

9. Which of the points $(0, 0)$, $(2, 1)$, and $(-1, 4)$ are solutions of $y < x + 2$?

10. For what values of y is $(2, y)$ a solution of $3x + y \leq 10$?

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Station 2 of 4: Boundary Lines and Shading

Name: _____ Date: _____

For each inequality, write the boundary line in slope-intercept (or vertical-line) form, say whether it is solid or dashed, and tell which side to shade.

1. $y > 3x - 2$

2. $y \leq -x + 5$

3. $y \geq \frac{1}{2}x$

4. $y < -4$

5. $x \geq 3$

6. $x < -1$

7. $2x + y > 6$

8. $3x - 2y \leq 8$

9. $x + 4y < 8$

10. $-x - y \geq 2$

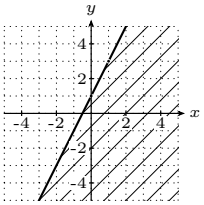
Algebra Review: Unit 4 — Graphing Inequalities on the Coordinate Plane

Station 3 of 4: Writing Inequalities from Graphs and Situations

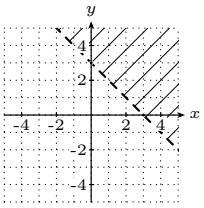
Name: _____ Date: _____

Write an inequality in two variables that matches each description. Diagrams are not drawn to scale.

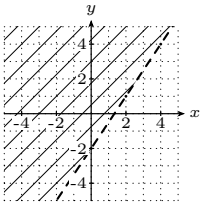
1. Write the inequality shown by the graph (the marked points lie on the boundary line).



2. Write the inequality shown by the graph (the marked points lie on the boundary line).

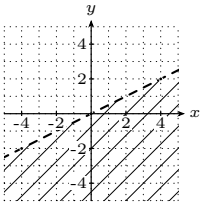


3. Write the inequality shown by the graph (the marked points lie on the boundary line).



4. Solid boundary line through $(0, 4)$ and $(2, 0)$; the region below is shaded.

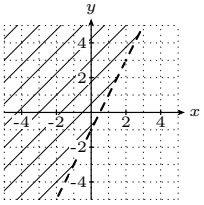
5. Write the inequality shown by the graph (the marked points lie on the boundary line).



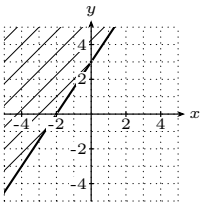
6. Solid vertical boundary line $x = -2$; the region to the right is shaded.

7. Solid horizontal boundary line $y = 5$; the region below is shaded.

8. Write the inequality shown by the graph (the marked points lie on the boundary line).



9. Write the inequality shown by the graph (the marked points lie on the boundary line).



10. Pens cost \$2 each (x) and notebooks cost \$5 each (y). You can spend at most \$60. Write an inequality, then decide whether 10 pens and 8 notebooks is possible.

Algebra Review: Unit 4 — Graphing Inequalities on the Coordinate Plane

Station 4 of 4: Systems of Linear Inequalities

Name: _____ Date: _____

Answer each question about the system. Show the work that supports your answer.

1. Is $(2, 3)$ a solution of $y > x$ and $y \leq 2x$?

2. Is $(-1, 1)$ a solution of $y \geq -x$ and $y < 2x + 3$?

3. Is $(4, 0)$ a solution of $x + y \leq 5$ and $x - y > 2$?

4. Is $(0, 0)$ a solution of $y > x - 2$ and $y < -x + 3$?

5. Find the vertex where the boundary lines $y = x + 1$ and $y = -x + 5$ meet.

6. Find the vertex where the boundary lines $y = 2x - 1$ and $y = -x + 8$ meet.

7. The region $x \geq 0$, $y \geq 0$, $x + y \leq 6$ is a triangle. Find its vertices.

8. The region $x \geq 1$, $x \leq 4$, $y \geq 0$, $y \leq 3$ is a rectangle. Find its vertices and its area.

9. Does the system $y > 2x + 1$ and $y < 2x - 3$ have any solutions? Explain.

10. A student earns \$15 per hour tutoring (x) and \$10 per hour at a café (y). She wants to earn at least \$120 but work no more than 15 hours. Write a system of inequalities, then decide whether $(4, 6)$ works.

Algebra Review: Unit 4 — Graphing Inequalities on the Coordinate Plane

Answer Key

Station 1: Testing Solutions of Linear Inequalities

1. Yes $3 > 1$ *true*
2. No $2 \leq 1$ *false*
3. Yes $5 \geq 5$ *true*
4. No $12 < 12$ *false*
5. Yes $4 \geq 4$ *true*
6. No $-3 < -3$ *false*
7. Yes $7 > 5$ *true*
8. $(3, -1)$ only $0 > 0$ *F;* $-1 > -3$ *T;* $2 > 2$ *F*
9. $(0, 0)$ and $(2, 1)$ $0 < 2$ *T;* $1 < 4$ *T;* $4 < 1$ *F*
10. $y \leq 4$

Station 2: Boundary Lines and Shading

1. $y = 3x - 2$; dashed; shade above
2. $y = -x + 5$; solid; shade below
3. $y = \frac{1}{2}x$; solid; shade above
4. $y = -4$; dashed; shade below
5. $x = 3$; solid; shade to the right
6. $x = -1$; dashed; shade to the left
7. $y = -2x + 6$; dashed; shade above $y > -2x + 6$
8. $y = \frac{3}{2}x - 4$; solid; shade above *dividing by* -2 *flips the sign:* $y \geq \frac{3}{2}x - 4$
9. $y = -\frac{1}{4}x + 2$; dashed; shade below
10. $y = -x - 2$; solid; shade below $y \leq -x - 2$

Station 3: Writing Inequalities from Graphs and Situations

1. $y \leq 2x + 1$
2. $y > -x + 3$
3. $y > \frac{3}{2}x - 2$
4. $y \leq -2x + 4$ *slope* $= \frac{0 - 4}{2 - 0} = -2$
5. $y < \frac{1}{2}x$
6. $x \geq -2$
7. $y \leq 5$
8. $y > 2x - 1$ *line:* $y = 2x - 1$*; origin:* $0 > -1$
9. $y \geq \frac{3}{2}x + 3$ *origin:* $0 < 3$ *is below, so shade above*
10. $2x + 5y \leq 60$; $20 + 40 = 60 \leq 60$, so yes

Station 4: Systems of Linear Inequalities

1. Yes $3 > 2$ *and* $3 \leq 4$
2. No $1 \geq 1$ *true, but* $1 < 1$ *false*
3. Yes $4 \leq 5$ *and* $4 > 2$
4. Yes $0 > -2$ *and* $0 < 3$
5. $(2, 3)$
6. $(3, 5)$
7. $(0, 0),\ (6, 0),\ (0, 6)$
8. $(1, 0),\ (4, 0),\ (4, 3),\ (1, 3)$; area 9
9. No solutions *the lines are parallel; a point cannot be above the higher line and below the lower line*
10. $15x + 10y \geq 120,\ x + y \leq 15,\ x \geq 0,\ y \geq 0$; $(4, 6)$ works: $120 \geq 120$ and $10 \leq 15$